Welcome to Remix 1.5.1

Your files are stored in indexedDB, 2.78 MB / 558.4 GB used

You can use this terminal to:

- Check transactions details and start debugging.
- Execute JavaScript scripts:
  - Input a script directly in the command line interface
  - Select a Javascript file in the file explorer and then run `remix.execute()` or `remix.exeCurrent()` in the command line interface
  - Right-click on a JavaScript file in the file explorer and then click `Run`

The following libraries are accessible:

- ethers.js

Type the library name to see available commands.

SolidityScan result for default_workspace/zykosv3/ZykosTokenV3_Toast.sol:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
1.	INCORRECT ACCESS CONTROL	critical	1	Access control plays an important role in the segregation smart contract. The contract is importing an access control library but the Lines: ["806:821","363:397","489:501","355:357
2.	UNCHECKED TRANSFER	high	2	Some tokens do not revert the transaction when the transf Lines: ["410:410","441:441","442:442","443:443
3.	PRECISION LOSS DURING DIVISION BY LARGE NUMBERS	medium	2	In Solidity, when dividing large numbers, precision loss c in inaccuracies or unexpected behaviors, especially when Lines: ["646:646"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
4.	LACK OF ZERO VALUE CHECK IN TOKEN TRANSFERS	low	1	The contract is found to lack a check for zero value transf disrupt the flow of the contract and lead to unexpected bel Lines: ["441:441","442:442","443:443","444:444
5.	MISSING ZERO ADDRESS VALIDATION	low	2	The contract is found to lack proper validation for zero ad such as having critical roles or operations reference the ze Lines: ["712:714","739:744","879:889","891:898
6.	OUTDATED COMPILER VERSION	low	2	Using an outdated compiler version can be problematic es Lines: ["2:2"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
7.	MISSING EVENTS	low	1	Events are inheritable members of contracts. When you ca These logs are associated with the address of the contract The contract was found to be missing these events on the Lines: ["712:714","739:744","806:821","879:889
8.	USE OF FLOATING PRAGMA	low	2	Solidity source files indicate the versions of the compiler The contract was found to be using a floating pragma whi Lines: ["2:2"]
9.	MISSING @DEV IN NATSPEC COMMENTS	informational	2	Natspec tags are missing from contract declarations in the

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
	FOR CONTRACT DECLARATION			Lines: ["30:39","41:915"]
10.	MISSING @PARAM IN NATSPEC COMMENTS FOR MODIFIERS	informational	2	Natspec comments are missing from a modifier declaratio Lines: ["251:254","256:259","261:264"]
11.	UNUSED RECEIVE FALLBACK	informational	2	The contract was found to be defining an empty function. It is not recommended to leave them empty unless there's Lines: ["912:914"]
12.	ADDING A RETURN STATEMENT WHEN THE FUNCTION DEFINES A NAMED RETURN VARIABLE IS REDUNDANT	informational	2	The contract defines a named return despite having a retur Lines: ["840:845","851:855","857:861"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
13.	MODIFIER CREATED BUT NEVER USED	informational	2	Solidity is a language that uses gas on deployment. Each u The contract was defining a modifier but none of the func Lines: ["261:264"]
14.	MISSING NATSPEC COMMENTS IN SCOPE BLOCKS	informational	2	NatSpec comments are missing from an unnamed scope b hard to understand why the scope block exists. Lines: ["333:346","334:337","390:392","421:423","521:5 75","723:725","757:759","796:798","798:800","8
15.	MISSING INDEXED KEYWORDS IN EVENTS	informational	2	Events are essential for tracking off-chain data and when t Lines: ["232:232","233:233","245:245"]
16.	VARIABLES SHOULD BE IMMUTABLE	informational	0	Constants and Immutables should be used in their appropr should only be used for literal values written into the code Lines: ["91:91","108:108"]
17.	USE SCIENTIFIC NOTATION	informational	0	Although the Solidity compiler can optimize exponentiati Solidity development. Lines: ["47:47","48:48","49:49","52:52","54:54
18	MISSING @NOTICE	informatio	2	Natspec comments are missing from a modifier declaratio

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
.	IN NATSPEC COMMENTS FOR MODIFIERS	nal		understand restrictions applied by modifiers. Lines: ["251:254","256:259","261:264"]
19.	MISSING @AUTHOR IN NATSPEC COMMENTS FOR CONTRACT DECLARATION	informational	2	Natspec .tags are missing from contract declarations in the verifying authenticity, and maintaining proper documenta Lines: ["30:39","41:915"]
20.	BLOCK VALUES AS A PROXY FOR TIME	informational	1	Contracts often need access to time values to perform cert For , as Ethereum block times are generally around 14 sec difficulty bomb. Due to variable block times, .should not be relied on for p Lines: ["344:344","369:369","402:402","387:387
21.	MISSING @NOTICE IN NATSPEC COMMENTS FOR FUNCTIONS	informational	2	Natspec .comments are missing from a function declaratio Lines: ["31:37","38:38","317:349","355:357","359:361", "823:834","840:845","847:849","851:855","857:8
22.	MISSING NATSPEC DESCRIPTIONS FOR PUBLIC VARIABLE DECLARATIONS	informational	2	Public variable declarations without NatSpec descriptions role and impact of these variables when interacting with t Lines: ["47:47","48:48","49:49","51:51","52:52","54:54 "104:104","107:107","108:108","124:124","125:1

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
				1″,″182:182″,″184:184″,″185:185″,″209:209″,″21
23.	MISSING @DEV IN NATSPEC COMMENTS FOR FUNCTIONS	informational	2	Natspec comments are missing from a function declaratio understand the function's implementation details. Lines: [″31:37″,″38:38″,″317:349″,″355:357″,″359:361″, ″823:834″,″840:845″,″847:849″,″851:855″,″857:8
24.	IF-STATEMENT REFACTORING	informational	0	In Solidity, we aim to write clear, efficient code that is bo to understand the trade-offs involved. Lines: [″796:800″]
25.	NAME MAPPING PARAMETERS	informational	0	After Solidity 0.8.18, a feature was introduced to name ma Lines: [″124:124″,″150:150″,″151:151″,″152:152
26.	AVOID ARITHMETIC DIRECTLY WITHIN ARRAY INDICES	informational	1	Using arithmetic directly within array indices () can introd variable for better readability and reliability. Lines: [″648:648″,″649:649″,″650:650″,″651:651

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
27.	MISSING @NOTICE IN NATSPEC COMMENTS FOR CONSTRUCTORS	informational	2	Natspec .comments are missing from a constructor declara Lines: ["270:311"]
28.	MISSING @INHERITDOC ON OVERRIDE FUNCTIONS	informational	2	Overridden functions in Solidity should include the .tag in understand the function's intended behavior. This lack of Lines: ["31:37","38:38","270:311","317:349","355:357", "806:821","823:834","840:845","847:849","851:85
29.	USE CALL INSTEAD OF TRANSFER OR SEND	informational	2	The contract was found to be using .or .function call. This Lines: ["455:455","456:456","457:457","458:458
30.	NAMED RETURN OF LOCAL VARIABLE SAVES GAS AS COMPARED TO RETURN STATEMENT	gas	0	The function having a return type is found to be declaring allocation. Lines: ["595:621","682:710","750:782"]
31.	CHEAPER CONDITIONAL OPERATORS	gas	0	During compilation, .is cheaper than .for unsigned integer Lines: ["364:364","374:374","407:407","400:400
32.	CHEAPER INEQUALITIES IN REQUIRE()	gas	1	The contract was found to be performing comparisons usi Lines: ["367:367","369:369","402:402","375:375

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
33.	SIMILAR DATATYPES CAN BE PACKED TOGETHER	gas	0	The contract is found to be using similar data types within gaps and increased gas consumption due to inefficient sto Lines: ["83:88","116:122","137:142","194:207"]
34.	CUSTOM ERRORS TO SAVE GAS	gas	2	The contract was found to be using statements. Since Sol This allows the developers to pass custom errors with dyn Lines: ["577:577"]
35.	VARIABLES DECLARED BUT NEVER USED	gas	0	The contract has declared a variable but it is not used anyw Unused variables increase the contract's size and complex Lines: ["47:47","60:60"]
36.	REVERTING FUNCTIONS CAN BE PAYABLE	gas	0	If a function modifier such as is used, the function will re provided. Lines: ["567:589","682:710","712:714","739:744
37.	AVOID RE-STORING VALUES	gas	0	The function is found to be allowing re-storing the value i

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
				old value and the new value are the same, not updating the Lines: ["879:889","891:898"]
38.	FUNCTIONS CAN BE IN-LINED	gas	0	The internal function was called only once throughout the Lines: ["317:349","435:447","449:462"]
39.	CACHE ADDRESS(THIS) WHEN USED MORE THAN ONCE	gas	0	The repeated usage of , within the contract could result in i Lines: ["305:305","375:375","408:408","381:381
40.	CHEAPER INEQUALITIES IN IF()	gas	1	The contract was found to be doing comparisons using ine When inside the : statements, non-strict inequalities , are us Lines: ["521:521","541:541","794:794","815:815
41.	GAS OPTIMIZATION IN INCREMENTS	gas	0	, costs less gas compared to , or , for unsigned integers. In , , Lines: ["333:333"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
42.	SPLITTING REQUIRE STATEMENTS	gas	1	Require statements when combined using operators in a si Lines: ["281:281","283:289","400:400","719:719
43.	PUBLIC CONSTANTS CAN BE PRIVATE	gas	2	Public constant variables cost more gas because the EVM Lines: ["47:47","48:48","49:49","51:51","52:52
44.	FUNCTION SHOULD RETURN STRUCT	gas	0	The function was detected to be returning multiple values. Consider using a .instead of multiple return values for the Lines: ["31:37","840:845","851:855","857:861"]
45.	DEFINE CONSTRUCTOR AS PAYABLE	gas	0	Developers can save around 10 opcodes and some gas if t However, it should be noted that it comes with risks becau Lines: ["270:311"]
46.	STORAGE VARIABLE CACHING IN MEMORY	gas	0	The contract is using the state variable multiple times in t .are expensive (100 gas after the 1st one) compared to ./.(3 Lines: ["270:311","363:397","399:429","512:561
47.	OPTIMIZING ADDRESS ID MAPPING	gas	0	Combining multiple address/ID mappings into a single ma It saves storage slot for the mapping and depending on the same storage slot. Lines: ["150:150","151:151","152:152","153:153
48.	UNNECESSARY CHECKED ARITHMETIC IN LOOP	gas	0	Increments inside a loop could never overflow due to the Lines: ["333:333","336:336"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
49.	GAS OPTIMIZATION FOR STATE VARIABLES	gas	2	Plus equals () costs more gas than addition operator. The s Lines: ["383:383","384:384","385:385","386:386
50.	EMIT USED IN LOOP	gas	0	In Solidity, when a code emits an event inside of a loop, i Lines: ["627:654"]
51.	ASSIGNING TO STRUCTS CAN BE MORE EFFICIENT	gas	0	The contract is found to contain a struct with multiple vari SLOAD and SSTORE operations happening at once Lines: ["339:345","530:535","550:555","695:706
52.	GAS INEFFICIENCY DUE TO MULTIPLE OPERANDS IN SINGLE IF/ELSEIF CONDITION	gas	2	The contract is found to use multiple operands within a si first condition fails, unless short-circuiting occurs, and the within loops. Lines: ["521:538","541:558","663:668","670:675
53.	USE SELFBALANCE() INSTEAD OF ADDRESS(THIS).BA	gas	0	In Solidity, efficient use of gas is paramount to ensure cos obtaining the contract's balance. Lines: ["905:905"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
	LANCE			
54.	DEFAULT INT VALUES ARE MANUALLY RESET	gas	0	The contract is found to inefficiently reset integer variable using the .keyword can achieve the same result while also Lines: ["611:611","630:630","644:644"]

Scan Summary:

Lines Analyzed: 732

Scan Score: 61.11

Issue Distribution: { "informational": 259, "low": 18, "medium": 1, "high": 6, "critical": 1, "gas": 134 }

For more details, go to SolidityScan.

SolidityScan result for default_workspace/ZykosTokenV3_FLAT.sol:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
1.	INCORRECT ACCESS CONTROL	critical	1	Access control plays an important role in the segregation contract. The contract is importing an access control library but th Lines: ["865:867","861:863","1227:1247","989:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
2.	INTEGER OVERFLOW IN UNCHECKED BLOCK	medium	0	The contract is found to potentially suffer from an intege allowing unexpected contract behavior or exploitation by Lines: ["1101:1101"]
3.	APPROVE FRONT-RUNNING ATTACK	medium	0	The method overrides the current allowance regardless o This can be abused by a token receiver when they try to v Meanwhile, if the sender decides to change the amount a front-running attack affecting the function. Lines: ["262:266","325:327","342:352"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
4.	RETURN VALUE OF LOW LEVEL CALLS	medium	2	The functions do not check the return value of low-level Lines: ["395:395","405:405","491:491"]
5.	MODIFIER SIDE EFFECTS	medium	0	Solidity functions should always use the Checks-Effects- Due to this reason, modifiers should only implement che The contract was found to be violating this pattern and th Lines: ["165:169"]
6.	PRECISION LOSS DURING DIVISION BY LARGE NUMBERS	medium	2	In Solidity, when dividing large numbers, precision loss inaccuracies or unexpected behaviors, especially when th Lines: ["1140:1140"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
7.	ACCOUNT EXISTENCE CHECK FOR LOW LEVEL CALLS	medium	0	The low-level calls such as the , , or , do not validate pric Lines: ["491:491"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
8.	UNCHECKED ARRAY LENGTH	medium	2	Ethereum is a very resource-constrained environment. Pr function exceeds the block gas limit, and transactions cal This becomes a security issue if an external actor influen E.g., if an array enumerates all registered addresses, an a Lines: ["1219:1219"]
9.	LACK OF ZERO VALUE CHECK IN TOKEN TRANSFERS	low	1	The contract is found to lack a check for zero value trans the flow of the contract and lead to unexpected behavior. Lines: ["912:912","945:945","946:946","947:947
10.	USE OWNABLE2STEP	low	0	.is safer than .for smart contracts because the owner cann Lines: ["73:73"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
11.	OUTDATED COMPILER VERSION	low	2	Using an outdated compiler version can be problematic e Lines: ["2:2"]
12.	MISSING ZERO ADDRESS VALIDATION	low	2	The contract is found to lack proper validation for zero a as having critical roles or operations reference the zero a Lines: ["65:69","82:85","87:90","376:385","387

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
13.	MISSING EVENTS	low	1	Events are inheritable members of contracts. When you c These logs are associated with the address of the contract The contract was found to be missing these events on the Lines: ["171:176","178:180","376:385","387:389
14.	CONSTRUCTORS SHOULD EMIT AN EVENT	informational	1	A constructor in a smart contract executes only once duri Lines: ["111:113","161:163","227:230"]
15.	INCORRECT NON-NEGATIVE CHECK	informational	2	The contract is found to have an incorrect non-negative c

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
	ON UNSIGNED INTEGER			greater than or equal to zero is redundant and unnecessar Lines: ["975:975"]
16.	CONTRACT NAME SHOULD USE PASCALCASE	informational	2	In Solidity, it's advisable to follow the established style g facilitates smoother collaboration among developers. By Lines: ["218:218","445:445"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
17.	AVOID ARITHMETIC DIRECTLY WITHIN ARRAY INDICES	informational	1	Using arithmetic directly within array indices () can intro for better readability and reliability. Lines: ["1144:1144","1145:1145","1146:1146","1
18.	IF-STATEMENT REFACTORING	informational	0	In Solidity, we aim to write clear, efficient code that is bo understand the trade-offs involved. Lines: ["298:306","425:429","433:440","1312:13

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
19.	MISSING @NOTICE IN NATSPEC COMMENTS FOR MODIFIERS	informational	2	Natspec comments are missing from a modifier declarat understand restrictions applied by modifiers. Lines: ["39:42","115:118","120:123","165:169",
20.	UNUSED RECEIVE FALLBACK	informational	2	The contract was found to be defining an empty function It is not recommended to leave them empty unless there' Lines: ["1451:1451"]
21.	INTERFACES SHOULD BE DECLARED IN A SEPARATE FILE	informational	1	Interfaces should be declared in separate files to maintai Lines: ["188:198","201:205","208:215"]
22.	MISSING NATSPEC DESCRIPTIONS FOR PUBLIC VARIABLE DECLARATIONS	informational	2	Public variable declarations without NatSpec description and impact of these variables when interacting with the c Lines:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
				["154:154","155:155","556:556","557:557","558: "602:602","603:603","609:609","610:610","611:6 684:684","685:685","686:686","687:687","689:68
23.	MISSING @NOTICE IN NATSPEC COMMENTS FOR CONSTRUCTORS	informational	2	Natspec comments are missing from a constructor declar Lines: ["32:37","111:113","161:163","227:230",
24.	MISSING NATSPEC COMMENTS IN ASSEMBLY BLOCKS	informational	2	NatSpec comments are missing from inline assembly () b auditors, maintainers, and developers to comprehend its f Lines: ["434:437"]
25.	MISSING @AUTHOR IN NATSPEC COMMENTS FOR ABSTRACT CONTRACTS	informational	2	Natspec tags are missing from abstract contract declarati Lines: ["9:21","24:70","73:99","102:150","153:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
26.	ADDING A RETURN STATEMENT WHEN THE FUNCTION DEFINES A NAMED RETURN VARIABLE IS REDUNDANT	informational	2	The contract defines a named return despite having a retu Lines: ["1359:1362","1368:1370","1372:1374"]
27.	IN-LINE ASSEMBLY DETECTED	informational	2	Inline assembly is a way to access the Ethereum Virtual Multiple vulnerabilities have been detected previously w Lines: ["434:437"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
28.	MISSING PAYABLE IN CALL FUNCTION	informational	0	The contract is using a method to make external calls alo Lines: ["1439:1439"]
29.	MISSING @DEV IN NATSPEC COMMENTS FOR ABSTRACT CONTRACTS	informational	2	Natspec tags are missing from abstract contract declarati Lines: ["9:21","24:70","73:99","102:150","153:
30.	MISSING @AUTHOR IN NATSPEC COMMENTS FOR CONTRACT DECLARATION	informational	2	Natspec tags are missing from contract declarations in th verifying authenticity, and maintaining proper documenta Lines: ["9:21","24:70","73:99","102:150","153:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
31.	INTERFACE NAMES SHOULD HAVE AN I AS THE FIRST CHARACTER	informational	1	The code defines an interface without the conventional ,p Lines: ["500:500"]
32.	MISSING @DEV IN NATSPEC COMMENTS FOR CONTRACT DECLARATION	informational	2	Natspec ,tags are missing from contract declarations in th Lines: ["9:21","24:70","73:99","102:150","153:
33.	REDUNDANT STATEMENTS	informational	0	The contract contains redundant statements where types Lines: ["837:852","1219:1224"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
34.	MISSING INDEXED KEYWORDS IN EVENTS	informational	2	Events are essential for tracking off-chain data and when Lines: ["748:748"]
35.	MISSING @PARAM IN NATSPEC COMMENTS FOR MODIFIERS	informational	2	Natspec comments are missing from a modifier declarati Lines: ["39:42","115:118","120:123","165:169",
36.	MISSING UNDERSCORE IN NAMING VARIABLES	informational	0	Solidity style guide suggests using underscores as the pre Lines: ["154:154","155:155","376:385","387:389

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
37.	MISSING NATSPEC COMMENTS IN SCOPE BLOCKS	informational	2	NatSpec comments are missing from an unnamed scope understand why the scope block exists. Lines: ["33:35","49:51","59:61","94:96","130:132","13 ","344:351","345:347","348:350","377:379","382 ,"1223:1223","885:890","893:895","925:927","89 71","1085:1106","1088:1091","1089:1089","1095: 62:1167","1165:1165","1169:1174","1172:1172", ","1438:1441","1442:1448","1445:1447"]
38.	MISSING @DEV IN NATSPEC COMMENTS FOR FUNCTIONS	informational	2	Natspec comments are missing from a function declarati understand the function's implementation details. Lines: ["10:12","14:16","18:20","44:46","48:52","54: :195","196:196","197:197","202:202","203:203", 340","342:352","357:365","366:366","367:367", 76","577:577","578:578","579:579","580:580", 080:1109","1111:1153","1159:1175","1181:1211", 8","1380:1382","1384:1386","1388:1390","1392:
39.	VARIABLES SHOULD BE IMMUTABLE	informational	0	Constants and Immutables should be used in their approp should only be used for literal values written into the co Lines: ["600:600","614:614"]
40.	MISSING @INHERITDOC ON OVERRIDE FUNCTIONS	informational	2	Overridden functions in Solidity should include the tag the function's intended behavior. This lack of clarity can Lines: ["10:12","14:16","18:20","32:37","44:46","48: 94:194","195:195","196:196","197:197","202:202 8:323","325:327","329:340","342:352","357:365 :493","501:507","576:576","577:577","578:578", 1001","1007:1052","1054:1074","1080:1109","111 1368:1370","1372:1374","1376:1378","1380:1382

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
41.	NAME MAPPING PARAMETERS	informational	0	After Solidity 0.8.18, a feature was introduced to name m Lines: ["630:630","655:655","656:656","657:657
42.	REVERT STATEMENTS WITHIN EXTERNAL AND PUBLIC FUNCTIONS CAN BE USED TO PERFORM DOS ATTACKS	informational	0	In public and external functions, improper use of .can be conditions from user input or contract state, an attacker c Lines: ["60:60","95:95","870:870","873:873","874:874" 33","1255:1255","1270:1270","1271:1271","1300:
43.	MODIFIER CREATED BUT NEVER USED	informational	2	Solidity is a language that uses gas on deployment. Each The contract was defining a modifier but none of the func Lines: ["765:768"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
44.	BLOCK VALUES AS A PROXY FOR TIME	informational	1	Contracts often need access to time values to perform cer For , as Ethereum block times are generally around 14 se bomb. Due to variable block times, should not be relied on for Lines: ["848:848","874:874","905:905","891:891
45.	MISSING @NOTICE IN NATSPEC COMMENTS FOR FUNCTIONS	informational	2	Natspec comments are missing from a function declarati Lines: ["10:12","14:16","18:20","44:46","48:52","54:5 :195","196:196","197:197","202:202","203:203", 340","342:352","357:365","366:366","367:367", 76","577:577","578:578","579:579","580:580","5 080:1109","1111:1153","1159:1175","1181:1211", 8","1380:1382","1384:1386","1388:1390","1392:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
46.	AVOID RE-STORING VALUES	gas	0	The function is found to be allowing re-storing the value value and the new value are the same, not updating the st Lines: ["65:69","82:85","329:340","1400:1412",
47.	OPTIMIZING ADDRESS ID MAPPING	gas	0	Combining multiple address/ID mappings into a single m It saves storage slot for the mapping and depending on th storage slot. Lines: ["219:219","220:220","655:655","656:656
48.	NAMED RETURN OF LOCAL VARIABLE SAVES GAS AS COMPARED TO RETURN STATEMENT	gas	0	The function having a return type is found to be declaring Lines: ["1080:1109","1181:1211","1269:1295"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
49.	ASSIGNING TO STRUCTS CAN BE MORE EFFICIENT	gas	0	The contract is found to contain a struct with multiple va and SSTORE operations happening at once Lines: ["843:849","1021:1026","1043:1048","119
50.	EMIT USED IN LOOP	gas	0	In Solidity, when a code emits an event inside of a loop, Lines: ["1115:1151"]
51.	SMALLER DATA TYPES COST MORE	gas	0	Usage of smaller integer types such as , , , or .in arithmet Lines: ["838:838","839:839"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
52.	CACHE ADDRESS(THIS) WHEN USED MORE THAN ONCE	gas	0	The repeated usage of .within the contract could result in Lines: ["377:377","378:378","393:393","392:392
53.	STORAGE VARIABLE CACHING IN MEMORY	gas	0	The contract is using the state variable multiple times in .are expensive (100 gas after the 1st one) compared to ./.(Lines: ["171:176","285:309","285:309","784:822","869: ","1181:1211","1254:1263","1269:1295","1297:13
54.	GAS INEFFICIENCY DUE TO MULTIPLE OPERANDS IN SINGLE IF/ELSEIF CONDITION	gas	2	The contract is found to use multiple operands within a s condition fails, unless short-circuiting occurs, and the co Lines: ["414:421","485:487","1010:1029","1031:

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
55.	SPLITTING REVERT STATEMENTS	gas	0	The contract is using multiple conditions in a single ,state Lines: ["485:487"]
56.	INTERNAL FUNCTIONS NEVER USED	gas	0	The contract declared internal functions but was not usin Since internal functions can only be called from inside th Lines: ["18:20","14:16","182:184","318:323","3
57.	REVERTING FUNCTIONS CAN BE PAYABLE	gas	0	If a function modifier such as ,is used, the function will r Lines: ["1054:1074","1181:1211","1213:1215","1

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
58.	USE SELFBALANCE() INSTEAD OF ADDRESS(THIS).BALANCE	gas	0	In Solidity, efficient use of gas is paramount to ensure co the contract's balance. Lines: ["377:377","392:392","1437:1437"]
59.	SPLITTING REQUIRE STATEMENTS	gas	1	Require statements when combined using operators in a s Lines: ["964:964"]
60.	DEFINE CONSTRUCTOR AS PAYABLE	gas	0	Developers can save around 10 opcodes and some gas if However, it should be noted that it comes with risks beca Lines: ["784:822"]
61.	FUNCTION SHOULD RETURN STRUCT	gas	0	The function was detected to be returning multiple values Consider using a instead of multiple return values for the Lines: ["501:507","1359:1362","1368:1370","137

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
62.	CHEAPER INEQUALITIES IN IF()	gas	1	The contract was found to be doing comparisons using in When inside the ,statements, non-strict inequalities ,are u Lines: ["377:377","392:392","433:433","873:873","874: 6:1346"]
63.	DEFAULT INT VALUES ARE MANUALLY RESET	gas	0	The contract is found to inefficiently reset integer variabl the .keyword can achieve the same result while also freei Lines: ["112:112","147:147","1097:1097","1120:
64.	SIMILAR DATATYPES CAN BE PACKED TOGETHER	gas	0	The contract is found to be using similar data types withi and increased gas consumption due to inefficient storage Lines: ["622:628","642:647","699:712"]

#	NAME	SEVERITY	CONFIDENCE	DESCRIPTION
65.	CHEAPER CONDITIONAL OPERATORS	gas	0	During compilation, .is cheaper than .for unsigned intege Lines: ["433:433"]
66.	FUNCTIONS CAN BE IN-LINED	gas	0	The internal function was called only once throughout th Lines: ["48:52","129:133","135:139","141:144",

Scan Summary:

Lines Analyzed: 1174

Scan Score: 61.50

Issue Distribution: { "informational": 663, "low": 33, "medium": 11, "high": 0, "critical": 1, "gas": 129 }

For more details, go to SolidityScan.

[block:75012064 txIndex:18]

from: 0xaE1...6B774

to: Address.(constructor)

value: 0 wei

data: 0x605...40033

logs: 0

hash: 0x72c...47447

[Verification] Contract deployed. Checking explorers for registration...

```
Etherscan verification skipped: API key not provided.

Please input the API key in Remix Settings - Connected Services OR Contract Verification Plugin Settings.

[Sourcify] Verification submitted. Awaiting confirmation...

[Sourcify] Verification Successful!  View Code
>
```